MSDS 629: Experiments in Data Science

Final Project

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Executive Summary

Through four experiments, we explore the impact on average browsing time of four factors: *Tile Size*, *Preview Delay*, *Preview Type*, and *Play Something*. This series of experiments include a 2^4 factorial experiment using all factors and three factorial experiments each with two factors, aiming to identify the factors with significant influence and the optimal configuration of these factors. We conclude that *Preview Type* and *Play Something* do not have significant effects on average browsing time, and the configuration that minimizes the metric of interest (MOI) is *Tile Size* at 25% and *Preview Delay* at 0.5 seconds. The expected average browsing time for this combination is 14.1975 minutes, with the 95% confidence interval being [14.0875, 14.3074].

Introduction

In the competitive world of streaming services, the smallest of changes in front-end UI can influence whether users can find a piece of media which catches their interest. This exhausting process of scrolling through seemingly infinite options can lead users to decision paralysis. This is a very real and important psychological phenomenon for Webflix, where if one can't make decisions to do an overwhelming number of decisions, they may get exhausted and simply give up. Webflix wants to ensure users can find a show or movie before they reach the point of giving up on engaging with their content.

To solve this problem, we engaged in a set of factorial experiments for factor screening, as well as to find optimal combinations of user-interface factors which may influence browsing time. The four factors we wanted to test for significance (both for main effects and interactive effects) included *Tile Size* (the height of the preview tile as a percentage of screen height), *Preview Delay* (the time in seconds between hovering the mouse over a tile and the start of a preview), *Preview Type* (the type of preview autoplayed, teaser/trailer vs. actual content), and *Play Something* (whether a button is on the screen which plays a random movie from a personalized collection). We hope to find the experimental condition which minimizes the length of time a user spends browsing.

In the remainder of the report, you will find a description of the experiments which were run and their results. The result of all of these will all be summarized in the conclusion at the end as well.

Experiments

Experiment 1: 2^4 factorial of all factors with extreme levels

The first experiment was a 2^4 factorial to investigate which of the four factors and their interaction terms, has a real, nonzero effect on the response. The response variable (RV) was browse time and the MOI was average browse time. No nuisance factors were named. The four factors and their levels are summarized in Table 1. Sample size was decided to be 110 (calculated with δ=1,σ=1.5, α=0.01, and power=0.99).

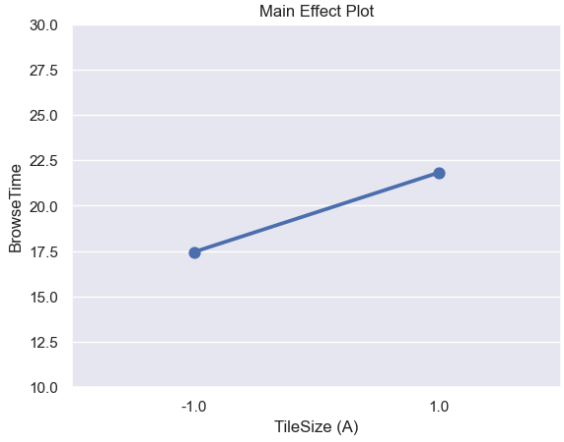
Table 1. Factors and levels for experiment 1

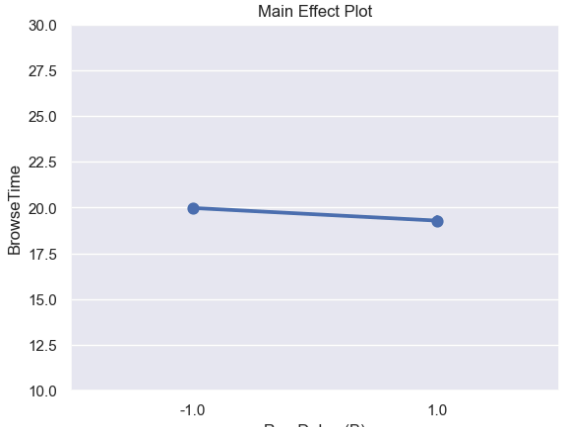
Factor	Low (-)	High (+)
Tile Size (x_1)	20	50
Preview Delay (x_2)	0	1
Preview Type (x_3)	TT	AC
Play Something (x_4)	N	Y

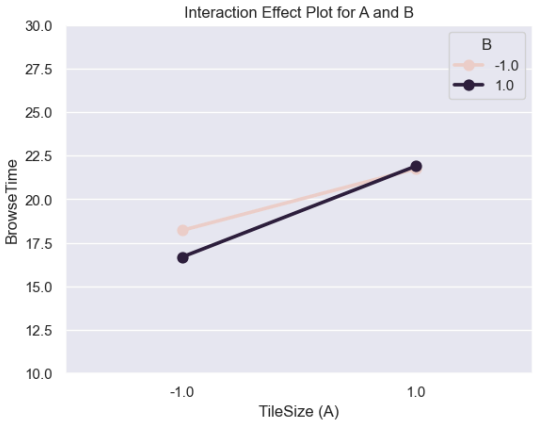
Using this data, we fitted a linear regression model with linear predictor:

$$\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_{12} x_1 x_2 + \beta_{13} x_1 x_3 + \beta_{14} x_1 x_4 + \beta_{23} x_2 x_3 + \beta_{24} x_2 x_4 + \beta_{34} x_3 x_4 + \beta_{123} x_1 x_2 x_3 + \beta_{124} x_1 x_2 x_4 + \beta_{134} x_1 x_3 x_4 + \beta_{234} x_2 x_3 x_4 + \beta_{1234} x_1 x_2 x_3 x_4$$

Below: Figure 1: Three Effect Plots involving Tile Size, Preview Delay and their Interaction







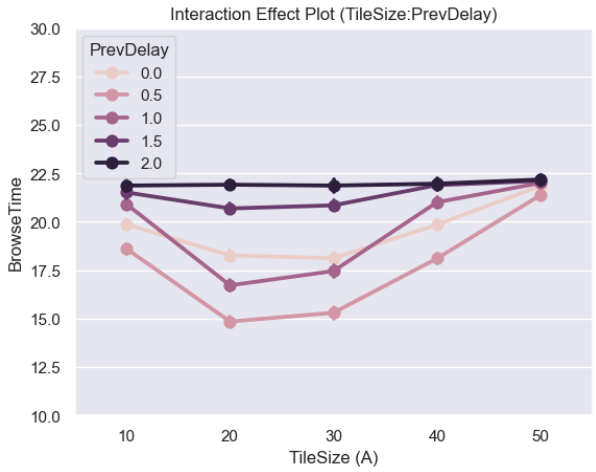
From the full model regression, the only statistically significant effects ($p < 0.05$) came from *Tile Size* ($P(T>|t|) \simeq 0.0$), *Preview Delay* ($P(T>|t|) \simeq 0.0$), and their interaction term *Tile Size:Preview Delay* ($P(T>|t|) \simeq 0.0$). Also notable was the interaction term *Tile Size:Play Something* ($P>|t| = 0.078$), which was close to the 5% significance cutoff. To avoid a potential Type II error, we

decided to test the significance of this interaction again in experiment 3.

Following this, partial F-tests were run which formally confirmed that no other main effects and interaction effects were significant (failed to reject H_0: $\beta_3 = \beta_{13} = \beta_{23} = \beta_{123} = \beta_4 = \beta_{24} = \beta_{124} = \beta_{134} = \beta_{34} = \beta_{234} = \beta_{1234} = 0$), and that the two main effects and one interaction term named above indeed were significant. We conclude that the factors *Tile Size* and *Preview Delay* have a significant impact on the MOI.

Experiment 2: Factorial for Tile Size:Preview Delay

Below: Figure 2: Interaction Effect Plot for Tile Size and Preview Delay



In the second experiment, we wanted to find the optimal levels for the two factors identified in the first experiment, *Tile Size* and *Preview Delay*, which were given 5 levels each: {10, 20, 30, 40, 50} and {0, 0.5, 1, 1.5, 2}, respectively. The other two factors were held at their default value. The RV was browse time and the MOI was average browse time. No nuisance factors were named. Sample size per condition was calculated to be 50 using updated σ=1 from the data gathered in experiment 1.

The interaction effect plot (Figure 2.) suggests that the optimal values for *Preview Delay* are either 0.5 or 1 and *Tile Size* is either 20 or 30.

After checking the distributions were roughly normal, Student's t-tests were run assuming equality of variances (F-tests of equal variances were performed to confirm this assumption), which showed that the conditions *Tile Size* = 20 and 30 with *Preview Delay* = 0.5 were not significantly different (the p-values were Bonferroni corrected). However, the t-tests also revealed that those two conditions led to significantly smaller average browsing times than any with *Preview Delay* = 1.

Since we cannot conclusively confirm the best condition out of the ones tested, we wanted to further test other levels for these *Tile Size* and *Preview Delay* in experiment 4. We also decided to update the sample size calculation for a smaller δ.

Experiment 3: 3x2 Factorial with *Tile Size* (3 levels) and *Play Something* (2 levels)

Our third experiment consists of a 3x2 factorial design with the following factors and their respective levels: *Tile Size* {20, 25, 30} and *Play Something* {Y, N}, while *Preview Delay* (0.5) and *Preview Type* (TT) were held constant. The primary goal in this experiment was to investigate whether the interaction *Tile Size:Play Something* significantly affects *Browse Time*, particularly within the chosen range of values for *Tile Size*.

We collected data for the 6 conditions, each with 200 units (more than the required sample size calculated using δ=0.5,σ=1, α=0.01, and power=0.99)

A partial F-test was performed to compare the full model with linear predictor $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_1 x_3 + \beta_5 x_2 x_3$ to the reduced model with linear predictor $\beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3$, with x_1, x_2, x_3 as defined in Table 1, effectively testing $H_0: \beta_4 = \beta_5$ *vs.* $H_A: \beta_4 \neq \beta_5$.

With p-value = 0.5842, we fail to reject the null hypothesis, concluding the interaction between *Tile Size* and *Play Something* has no significant effect on *Browse Time* at the 5% significance level.

Experiment 4: 5x3 Factorial Design with *Tile Size* (5 levels) and *Preview Delay* (3 levels)

With *Tile Size* and *Preview Delay* commanding significant effects on *Browse Time* (from experiment 1), our aim for our fourth and final experiment was to perform a grid search with nuanced values to identify the most optimal configuration to minimize *Browse Time*. We used a study design with 15 total conditions using two factors, *Tile Size* with levels {20, 22.5, 25, 27.5, 30} and *Preview Delay* with levels {0.3, 0.5, 0.7}, holding other two factors constant and using n=300 sample size (more than the required sample size calculated using δ=0.5, σ=1, FWER=0.01, power=0.99).

F-test of overall significance shows that average browse times is not the same across levels of *Tile Size* and levels *Preview Delay*. Looking at average browse times across all 15 conditions, we see that Condition 8: (*Tile Size* = 25, *Preview Delay* = 0.5) results in the lowest average *Browse Time* of 14.1975 minutes. Welch's t-tests were performed to test against the 14 remaining conditions using Holm's correction method at the 1% significance level. All 14 null hypotheses were rejected, concluding that *Tile Size* = 25 and *Preview Delay* = 0.5 is the optimal configuration for minimizing the MOI.

	TileSize	PrevDelay	BrowseTime
0	20.0	0.3	15.314084
1	20.0	0.5	14.902439
2	20.0	0.7	15.988914
3	22.5	0.3	14.753291
4	22.5	0.5	14.546994
5	22.5	0.7	15.167185
6	25.0	0.3	14.947889
7	25.0	0.5	14.197461
8	25.0	0.7	14.456259
9	27.5	0.3	15.027190
10	27.5	0.5	14.394954
11	27.5	0.7	14.849191
12	30.0	0.3	15.085146
13	30.0	0.5	15.145954
14	30.0	0.7	15.944367

Figure 3: Average Browse Time for each Condition (0 indexed)

1. $H_0 : \mu_8 \geq \mu_1$ vs. $H_A : \mu_8 < \mu_1$
2. $H_0 : \mu_8 \geq \mu_2$ vs. $H_A : \mu_8 < \mu_2$
3. $H_0 : \mu_8 \geq \mu_3$ vs. $H_A : \mu_8 < \mu_3$
4. $H_0 : \mu_8 \geq \mu_4$ vs. $H_A : \mu_8 < \mu_4$
5. $H_0 : \mu_8 \geq \mu_5$ vs. $H_A : \mu_8 < \mu_5$
6. $H_0 : \mu_8 \geq \mu_6$ vs. $H_A : \mu_8 < \mu_6$
7. $H_0 : \mu_8 \geq \mu_7$ vs. $H_A : \mu_8 < \mu_7$
8. $H_0 : \mu_8 \geq \mu_9$ vs. $H_A : \mu_8 < \mu_9$
9. $H_0 : \mu_8 \geq \mu_{10}$ vs. $H_A : \mu_8 < \mu_{10}$
10. $H_0 : \mu_8 \geq \mu_{11}$ vs. $H_A : \mu_8 < \mu_{11}$
11. $H_0 : \mu_8 \geq \mu_{12}$ vs. $H_A : \mu_8 < \mu_{12}$
12. $H_0 : \mu_8 \geq \mu_{13}$ vs. $H_A : \mu_8 < \mu_{13}$
13. $H_0 : \mu_8 \geq \mu_{14}$ vs. $H_A : \mu_8 < \mu_{14}$
14. $H_0 : \mu_8 \geq \mu_{15}$ vs. $H_A : \mu_8 < \mu_{15}$

Figure 4: Hypotheses Tests Against Cond8 (25, 0.5)

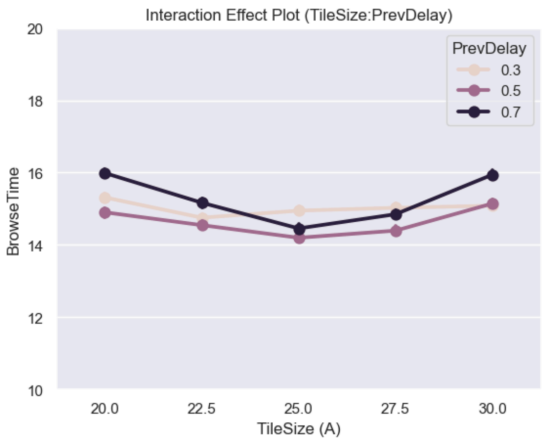


Figure 5: Interaction Effect Plot for Tile Size by and Preview Delay

Conclusion

Webflix users experience decision paralysis when overwhelmed with viewing options, leading to extended browsing times and potential disengagement. To address this, we conducted four factorial experiments testing how *Tile Size, Preview Delay, Preview Type*, and *Play Something* influence browsing behavior. Our initial 2^4 factorial experiment (Experiment 1) revealed that only two of the four factors significantly influence browsing time: *Tile Size* and *Preview Delay*. Both factors demonstrated strong main effects along with a significant interaction effect, indicating their combined influence on average browse time. Neither *Preview Type* nor *Play Something* demonstrated a significant main or interaction effect on browsing time, allowing us to eliminate these factors from further analysis. Subsequent experiments focused on narrowing the optimal region for *Tile Size* and *Preview Delay*. A fine-grained grid search over five *Tile Size* levels and three *Preview Delay* levels identified the optimal configuration.

We recommend Webflix implement *Tile Size* at 25% with a 0.5 second *Preview Delay*. *Preview Type* and *Play Something* can remain at their default settings, as they do not significantly affect browsing time. Our findings suggest this configuration is optimal for reducing user browsing time and improving the overall streaming experience, with an expected average browsing time of 14.1975 minutes (95% confidence interval: [14.0875, 14.3074]).

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